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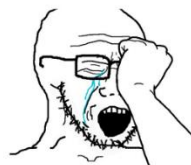
Optimization 2 : Towards Gradient Descent.

When your friend asks what the normal vector to a plane looks like



Agenda for today :

- Recap
- Derivatives in a single dimension.
- Derivatives in multiple dimensions.

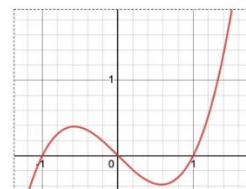


Real Analysis Student

YOU NEED THAT FOR $f: A \rightarrow \mathbb{R}$, $c \in A$, THE FUNCTION IS CONTINUOUS AT c IF AND ONLY IF $\forall \epsilon > 0 \exists \delta > 0 \ni |x-c| < \delta$ and $x \in A$ implies $|f(x)-f(c)| < \epsilon!!!$ OTHERWISE IT'S NOT SUFFICIENTLY RIGOROUS!!!!



Precalculus Student



If I can draw it without picking my pen up, it's continuous.

Q. What is the value of x at which this maximum value occurs?

- it is $x=2$ as we can see from the plot

Detailed Recap:

$$w_0^*, \bar{w}^* = \arg \max_{\bar{w}, w_0} f(\theta, \bar{w}, w_0)$$

→ Simple Example (1-0)

$$x^* = \arg \max_x \underbrace{-(x-2)^2}_{\text{max value is 0}}$$

↓
2

but, x which gives us the
max value $\Rightarrow x=2$

Cheat Sheet

(1) Left Hand Limit

$$\lim_{x \rightarrow a^-} f(x)$$

(2) Right Hand Limit

$$\lim_{x \rightarrow a^+} f(x)$$

(3) Domain set of all possible inputs

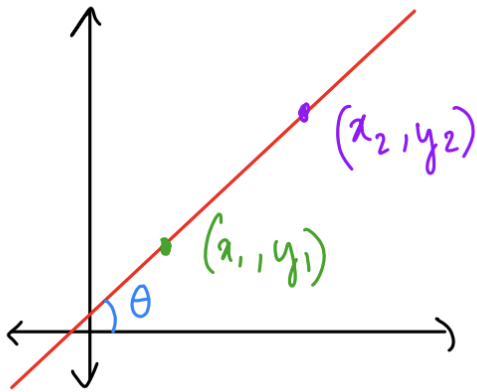
(4) Range set of all possible outputs

Differentiation geometric picture

Let's say we have 2 points on the line (x_1, y_1) and (x_2, y_2)

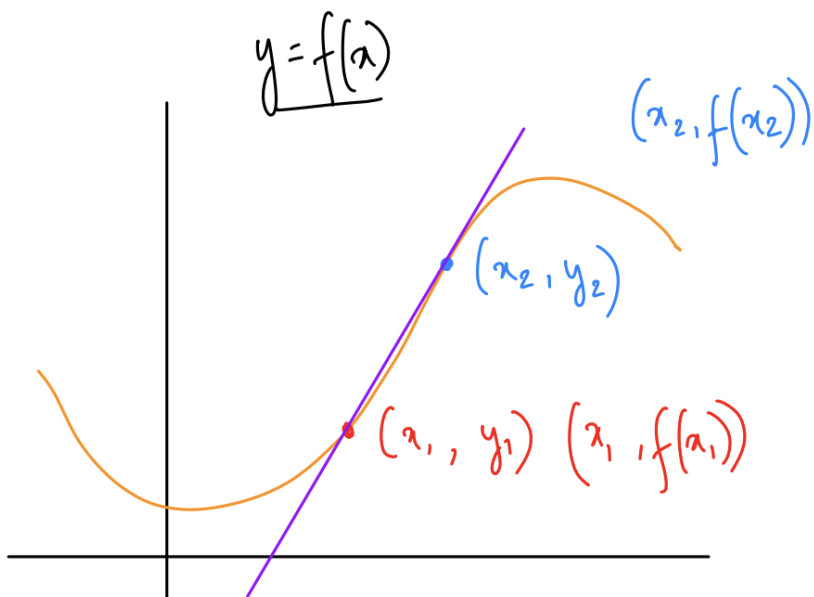
Q. What is the slope of this line?

Differentiation: Geometric Picture.



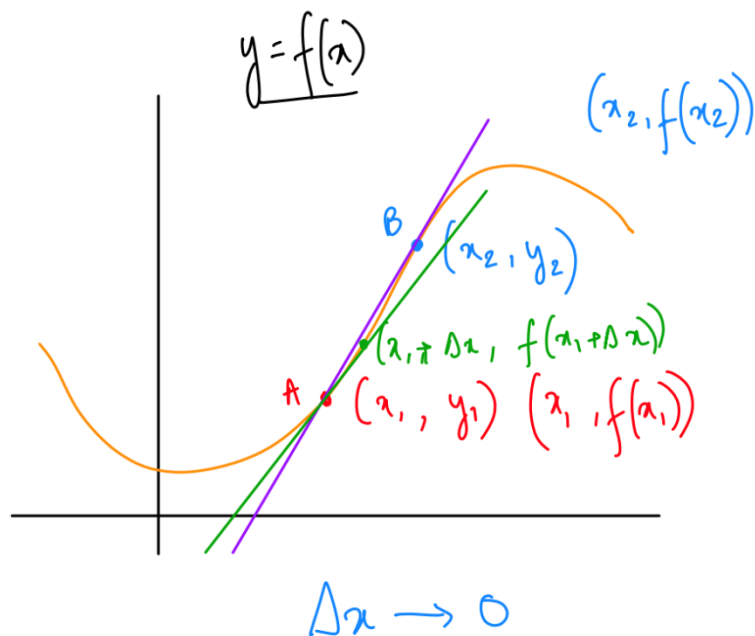
$$m = \tan \theta$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$



$$(x + \Delta x)$$

$$3 + 0.000000001$$

$$m = \frac{f(x_1 + \Delta x) - f(x_1)}{x_1 + \Delta x - x_1}$$

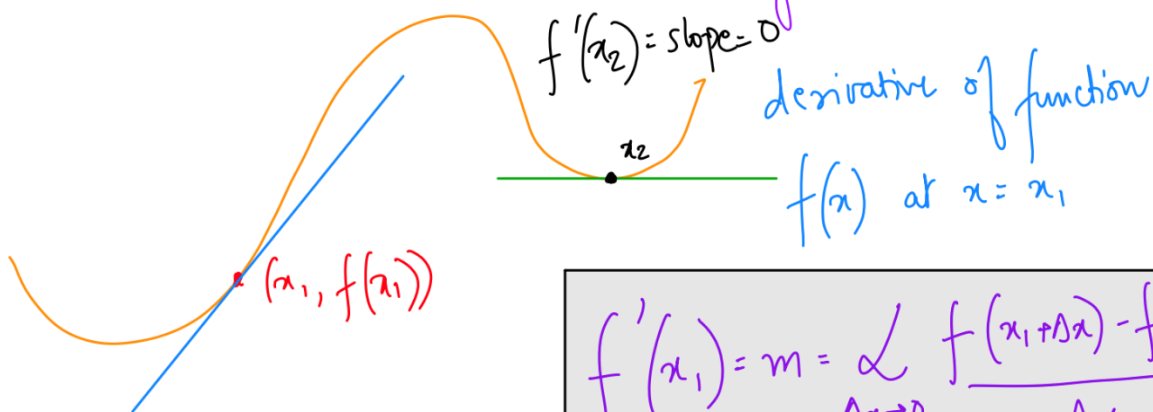
$$m = \frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$$

Terminologies

- If a line joins any 2 points on a function curve we call that as a **secant**
- If a line touches only 1 points on a function or curve we call that as a **Tangent**

* If a line joins 2 points on a function curve, it is called as "Secant".

* If a line touches 1 point on a function curve, it is called as "Tangent".



$$f'(x_1) = m = \lim_{\Delta x \rightarrow 0} \frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$$

Note:

- Δx is also represented as δx at many places but they refer to the same thing in this context

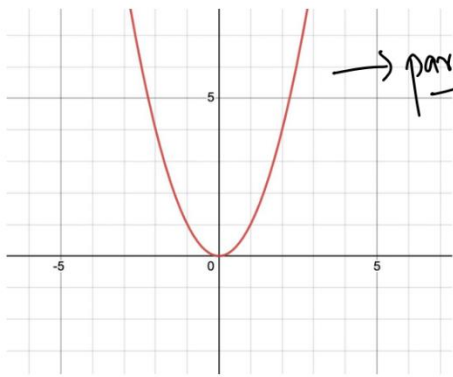
How to find derivative for any given function

How to find derivative function for any given function $f(x)$?

- ① Simplest way - directly using the definition
"Ab initio method".

$$\frac{d}{dx} f(x) = f'(x_1) = \lim_{\Delta x \rightarrow 0} \frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$$

- ② First, we will remember derivatives of some well known functions and then we will use Rules to find derivative of "Unknown" function.



→ parabola

$$f(x) = x^2$$

$$\frac{d}{dx} f(x) = f'(x)$$

$$\frac{d}{dx} f(x) = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(x+\Delta x)^2 - x^2}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\cancel{x^2} + (\Delta x)^2 + 2x\Delta x - \cancel{x^2}}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\cancel{\Delta x} (\Delta x + 2x)}{\cancel{\Delta x}}$$

$$= \lim_{\Delta x \rightarrow 0} 2x + \Delta x \overset{0}{\rightarrow} = 2x$$

$$\frac{d}{dx} f(x) = 2x$$

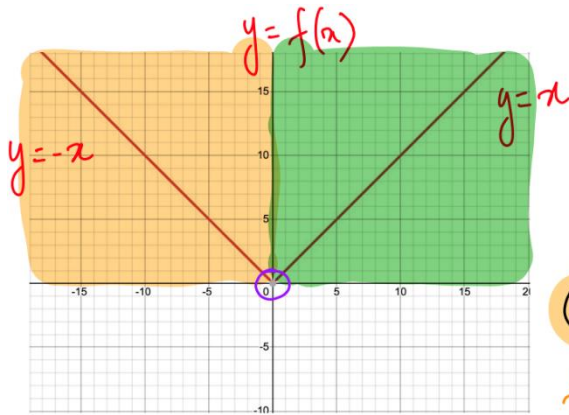
$$f(x) = x^2$$

$$\frac{d}{dx} x^n = n \cdot x^{n-1}$$

$$\frac{d}{dx} x^2 = 2x^{2-1}$$

$$= 2x^1 = 2x$$

↳ Very Useful later on



$$f(x) = \begin{cases} x & x > 0 \\ -x & x < 0 \\ 0 & x = 0 \end{cases}$$

Case 1: $x > 0$

$$f(x) = x$$

$$f'(x) = 1$$

Case 2: $x < 0$

$$f(x) = -x$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{-(x + \Delta x) - (-x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{-x - \Delta x + x}{\Delta x} = \lim_{\Delta x \rightarrow 0} -\frac{\Delta x}{\Delta x} = -1$$

Case 3: $x = 0$

$$\lim_{x \rightarrow 0^+} f'(x) = 1$$

$$\lim_{x \rightarrow 0^-} f'(x) = -1$$

Not differentiable

How to check if a function is differentiable

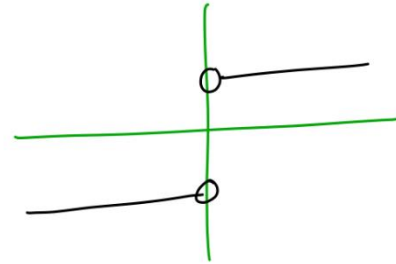
Steps

1. we will find out the value of first derivative of the function ie $f'(x)$
2. Find out if the values are continuous or discontinuous at $f'(x)$
 - For this we have to create a plot of $f'(x)$ and then check whether it is continuous or discontinuous
 - **So if $f'(x)$ is discontinuous then we say that $f(x)$ is not differentiable**

How do we check if a function is differentiable?

① Step 1: Get $f'(x)$

② Step 2:



If $f'(x)$ is discontinuous, we say
 $f(x)$ is not differentiable

checking differentiability for $f(x)=|x|$

Let's understand the same concept visually via the tangents concept and checking of function $f(x)=|x|$ is differentiable or not

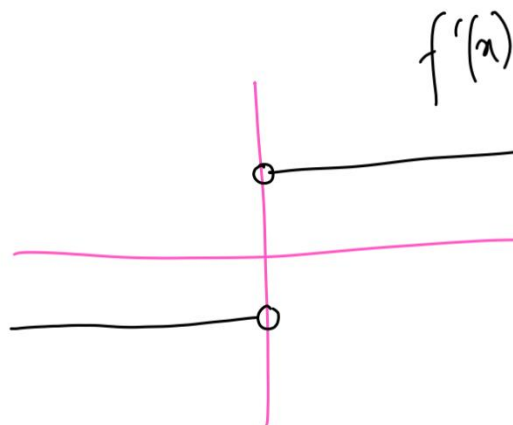
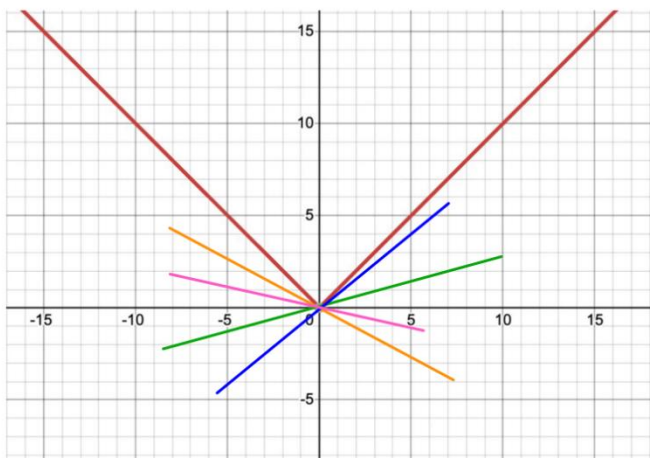
When we say that a function is differentiable we say that we can find out the slope of the tangent

Q. Now for the given function $f(x)=|x|$ at point $x=0$ can we find the slope?

- Your first question should be how should we create this tangent as you can see that there can be multiple tangents possible at this point.
- Because of this we say that at point $x=0$ we cannot differentiate the function $f(x)=|x|$

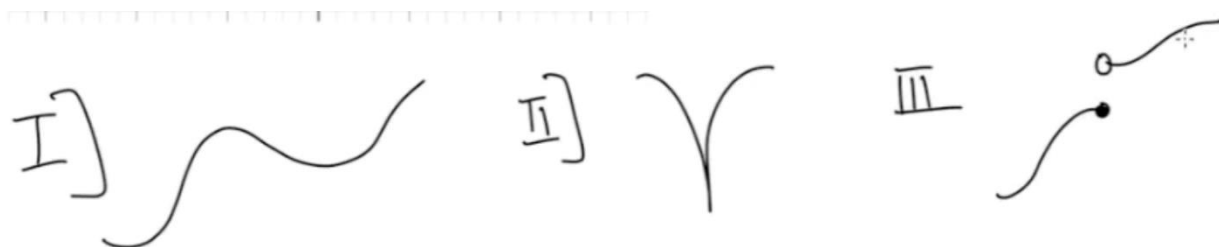
Mathematically we'll do something as follows based on the above mentioned steps

- We'll first take the first derivative of the function $f(x)$
 - it'll be +1 where values of x is greater than 0
 - it'll be -1 where values of x is less than 0
 - And we cannot differentiate at $x=0$
- Thus we say that the function $f'(x)$ is discontinuous and hence the function $f(x)$ is not differentiable

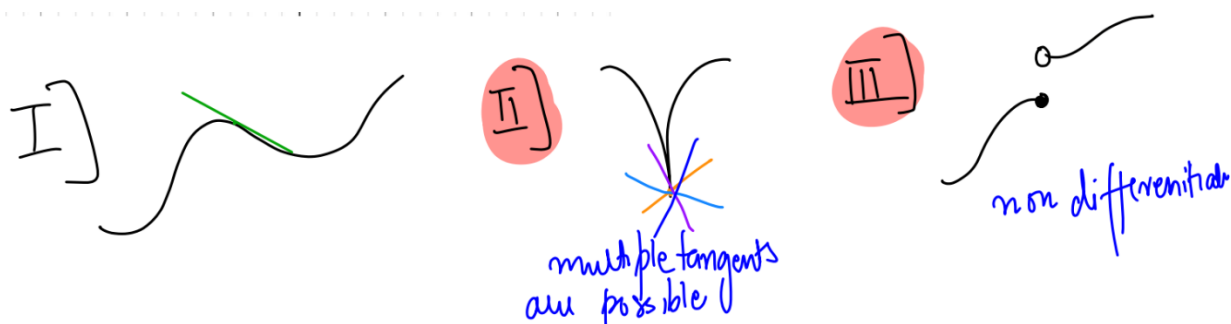


Q. From the given below function identify the function which is non differentiable.

- Note that these are $f(x)$ and not $f'(x)$



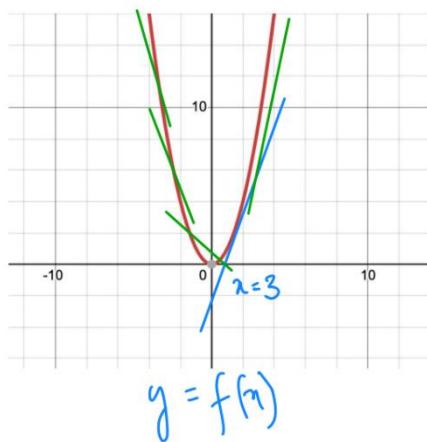
- The correct answer is 2 and 3
 - for function 2 we can have multiple tangents at the point where the function is dipping
 - for function 3 is itself a discontinuous function hence it cannot be differentiated



checking differentiability for $f(x) = x^2$

Instructor note:

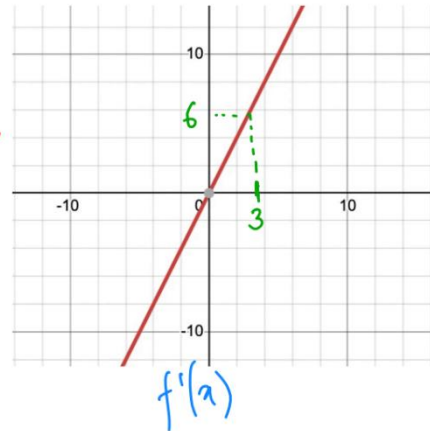
- Follow the same steps to prove that this function is differentiable



$$y = f(x) = x^2$$

$$y' = f'(x) = 2x$$

$$y' = f'(x=3) = 6$$



in the plot $f'(x)$ we are essentially tracking what is the slope for different values of x and this is how the plot is generated in the first place

Some commonly used derivatives.

$$(a) \quad \frac{d}{dx} x^n = nx^{n-1}$$

$$(e) \quad \frac{d}{dx} \sin(x) = \cos(x)$$

$$(b) \quad \frac{d}{dx} \log(x) = \frac{1}{x}$$

$$(f) \quad \frac{d}{dx} \cos(x) = -\sin(x)$$

$$(c) \quad \frac{d}{dx} e^x = e^x$$

$$(g) \quad \frac{d}{dx} \tan(x) = \sec^2(x)$$

$$(d) \quad \frac{d}{dx} c = 0$$

Rules of differentiation

Linearity rule

Rules of differentiation:

(1) Linearity Rule:

eg: $f(x) = x^3 + \ln x$
 $f'(x) = 3x^2 + 1/x$

$$(a) \quad h(x) = g(x) + f(x)$$
$$h'(x) = g'(x) + f'(x)$$

eg $f(x) = 3 \sin(x)$
 $f'(x) = 3 \cos(x)$

$$(b) \quad h(x) = c \cdot f(x) \quad c \in \mathbb{R}$$
$$h'(x) = c f'(x)$$

Product rule

(2) Product Rule:

eg $h(x) = \underline{x} \underline{\sin x}$

$$h'(x) = f'(x) \cdot g(x) + f(x) g'(x)$$

$$h'(x) = \sin x + x \cos(x)$$

$$h(x) = \sin(x) \cdot \cos(x)$$

$$h'(x) = \cos(x) \cos(x) + \sin(x) (-\sin(x))$$
$$= \cos^2(x) - \sin^2(x) = \cos(2x)$$

Quotient rule

③ Quotient Rule

$$h(x) = \frac{f(x)}{g(x)}$$

$$\text{eg } f(x) = \frac{\sin(x)}{\cos(x)}$$

$$f'(x) = \frac{\cos(x)\cos(x) - (\sin(x) \times (-\sin(x)))}{[\cos(x)]^2}$$

$$= \frac{\cos^2(x) - (-\sin^2(x))}{[\cos(x)]^2}$$

$$= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)} = \sec^2(x)$$

$$\frac{d}{dx}(h(x)) = h'(x)$$

$$h'(x) = \frac{g(x) \cdot f'(x) - g'(x) \cdot f(x)}{[g(x)]^2}$$

Chain rule

④ Chain Rule

$$h(x) = f(g(x))$$

$$h'(x) = f'(g(x)) \cdot g'(x)$$

$$\text{eg } h(x) = \log(x^2)$$

$$h'(x) = \frac{1}{x} \cdot 2x$$

$$h'(x) = \frac{2}{x}$$

$$h(x) = e^{-x}$$

$$h'(x) = e^{-x} \cdot -1$$

$$h'(x) = -e^{-x}$$

How do we use derivatives for optimization

For the given curve

- all the points to the left of point 1 will have a negative slope
- similarly for point 3 and 5 we'll have a negative slope for all the points to the left of point 3 and 5

And

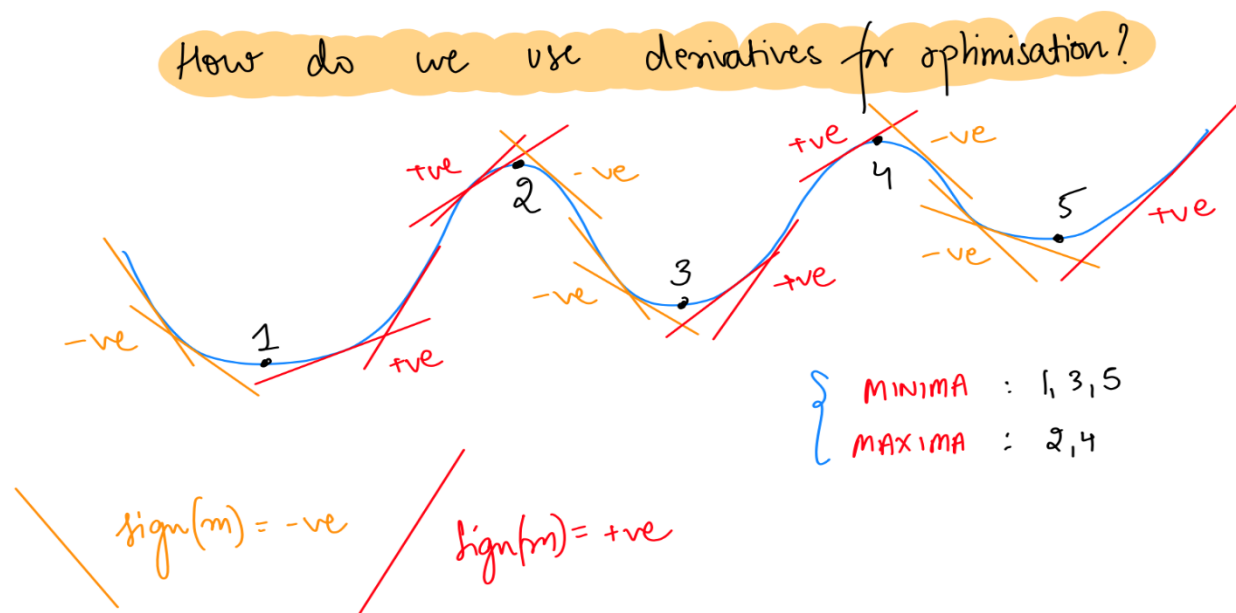
- all the points to the right of point 1 will have a positive slope
- similarly for point 3 and 5 we'll have a positive slope for all the points to the right of point 3 and 5

For points 2 and 4

- on the left side we'll have a positive slope and on the right side we'll have negative slopes for all the points

Now we call all this 5 points as **critical points** where the sign of the slope is changing

- For points 1, 3 and 5 we can observe that the slope is changing from negative to positive and we call this points as **Minima**
- For points 2 and 4 we can observe that the slope is changing from positive to negative and we call this points as **Maxima**



How to identify critical points for any function

Q. for the above function what is the slope or derivative at point 1,2,3,4 and 5?

- It is 0

Q. Out of points 1,3 and 5 which point is the lowest?

- it is point 1 we call this point as the **global minima** and the points 3 and 5 are called as the **local minima** of the function

Q. Out of points 2 and 4 which point has the highest peak or value?

- it is point 4 we call this point as the **global maxima** and the points 2 is called as the **local maxima** of the function

So to identify critical points of any given function we find the point where derivative is equal to 0 and at that point the function would be flat

① How to find out Candidate points for minima & maxima?

Given $f(x)$, check $f'(x) = 0$

"if derivative is 0, then the func is approximately flat at that point."

How to identify if a critical point is minima or maxima

One way to is as follows:

- let say we want to identify if point 1 is minima or maxima what we can do is we take another point which very close and left of point 1 and find the slope
- and similarly we take another point which is very close and to the right side of point 1 and find the slope
- next we verify if the change in slope if from negative to positive from left to right then it is minima and if the change in slope is from positive to negative then it is a maxima

But this is a tedious and a length process

Second way is to take a second derivate at all those critical points

Now the second derivate can give us 3 possible values

1. $f''(x) > 0$ if the second derivative is positive then we say that we have reached a **minima**

2. $f''(x) < 0$ ie the second derivative is negative then we say that we have reached a **maxima**
3. $f''(x) = 0$ ie the second derivative is positive then we say that we have reached a **Saddle point**

When we looked at the plot of the first derivative of the slope we learnt that it is nothing but the slope at different values of x now when we look at the second derivative essentially what we are finding out is the **rate of change of slopes**

② Once we have identified candidate points, how will we check for minima & maxima?

Given $f(x)$, Candidate points

Calculate $f''(x)$ Rate of change of slopes

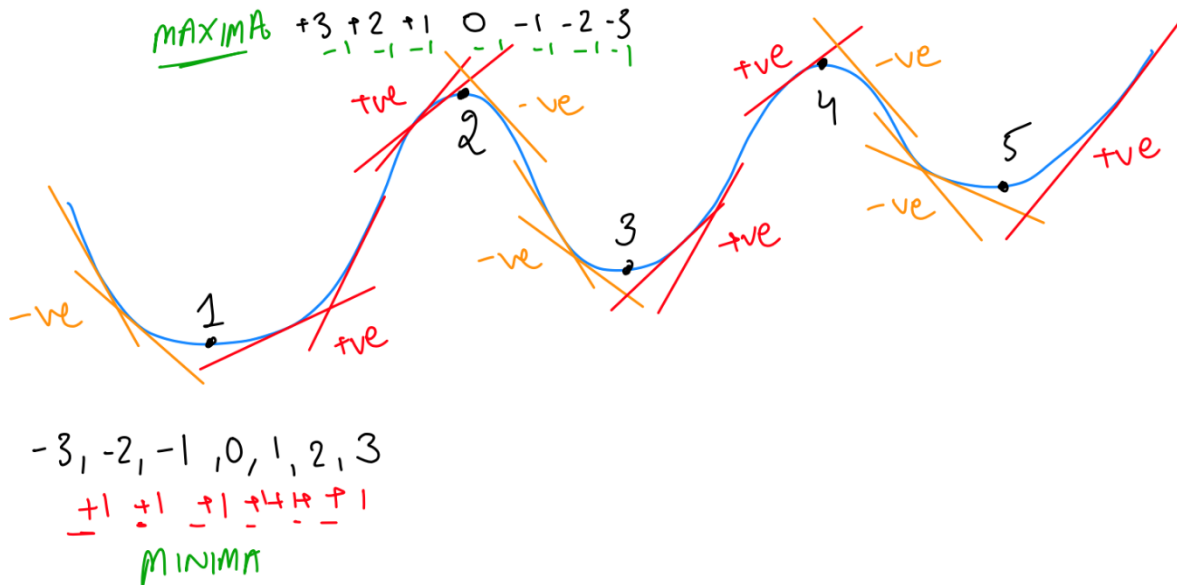
$f''(x) > 0$	MINIMA
$f''(x) < 0$	MAXIMA
$f''(x) = 0$	Saddle point.

Let's analyze the curve that we saw earlier for point 1

- For point 1 we saw that the slopes is going from negative to positive ie for example from -3, -2, -1, 0, 1, 2, 3
- Now to go from -3 to -2 we are adding a 1 to it so essentially we are changing the slope by adding a value similarly for rest of the points
- So we can say that there is a positive change of rate of slopes from -3 to 3 which means that the double derivate is positive
- **Thus when there is a positive rate of change in slopes then it will be a minima**

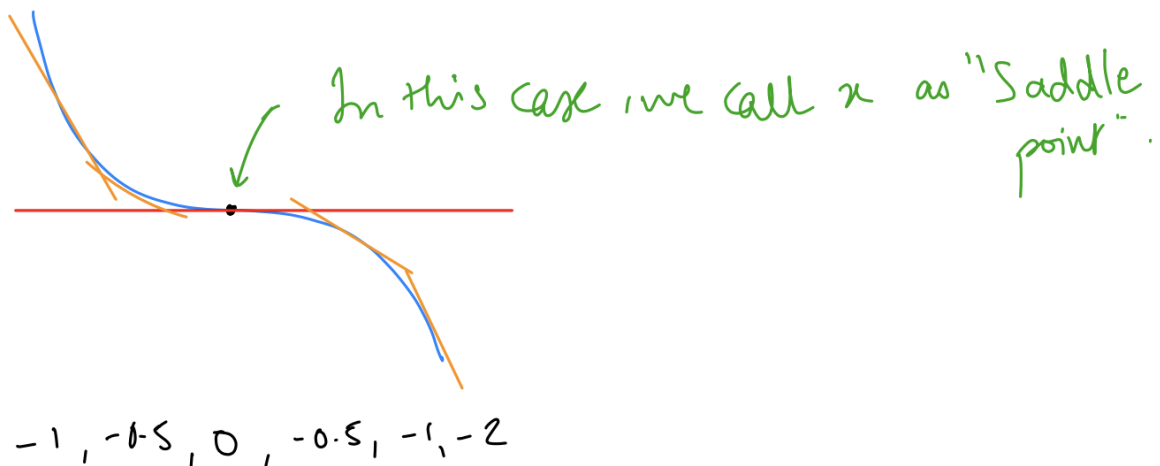
Let's analyze the curve that we saw earlier for point 2

- For point 2 we saw that the slopes is going from positive to negative ie for example from 3, 2, 1, 0, -1, -2, -3
- Now to go from 3 to 2 we are subtracting by -1 to it so essentially we are changing the slope by subtracting a value similarly for rest of the points.
- So we can say that there is a negative change of rate of slopes from 3 to -3 which means that the double derivate is negative
- **Thus when there is a negative rate of change in slopes then it will be a maxima**

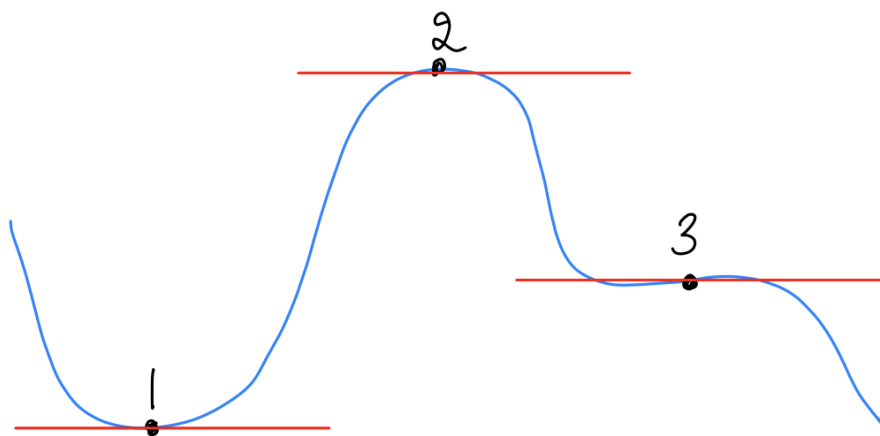


What is a Saddle point?

- Let say we have a curve as shown
- For the marked point do you see that the slope is 0
- Points to the left of this point will have a negative slope and points to the right side will also have a negative slope thus the signs are not changing
- We call this points as **Saddle points**
- These are points where the derivative is 0 but they are neither maxima or minima



For the given function determines whether point 1, 2 and 3 are minima, maxima or a saddle point



$$y = f(x)$$

$$f'(x) = 0$$

x_1 MINIMA

x_2 MAXIMA

x_3 SADDLE
POINT.